

Gender similarities and differences in early childhood development in low- and middle-income countries

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Abstract

Gender differences in early childhood development are under-researched in low- and middle-income countries (LMICs) where almost 90% of the world's young children live. This study examines the direction, magnitude, and variation of gender similarities and differences in 3- and 4-year-olds' early childhood development in 71 low- and middle-income countries (LMICs). Moreover, we explore whether country-level gender inequality correlates with gender differences in early development. We analyzed Early Childhood Development Index (ECDI) data from UNICEF's Multiple Indicators Cluster Surveys and the Demographic and Health Surveys between 2010 and 2019 ($n = 226,980$). Using logistic regression, we estimated children's odds of being developmentally on track in literacy-numeracy, approaches to learning, social-emotional, and physical domains. We also used meta-analytic regression to examine associations between early childhood development and country-level indicators of gender inequality. In approximately one-half of LMICs, odds of being developmentally on track did not significantly differ for girls and boys; in countries with significant differences, girls were more likely to be on track than boys (odds ratio [OR] = 1.21, 95% confidence interval [CI] = [1.18–1.24], $p < .001$). Differences favoring girls were largest in the social-emotional domain (OR = 1.27, 95% CI = [1.24–1.30], $p < .001$). Country-level gender development index explained a substantial amount of variation in gender differences across countries. Overall, findings suggest that when gender differences exist, girls are more likely than boys to be developmentally on track, particularly in the social-emotional domain and in countries with greater gender equality.

Keywords

Gender differences, childhood development, low- and middle-income countries, early literacy, early numeracy, social-emotional skills

The first 5 years of life constitute a critical period of human development where brain and foundational skills develop in response to reciprocal interactions and environmental inputs (Black et al., 2017). Children's sex and gender categories assigned at birth often determine the nature and quality of social interactions and investment decisions families make for them (Baker & Milligan, 2016; Barcellos et al., 2014; Raley & Blanchi, 2006). As infant and toddler brains have evolved to detect the subtlest nuances in their environments, even small gender differences in early experiences could cascade into children's developmental trajectories and later-life outcomes (Berens & Nelson, 2019; Martin & Ruble, 2004). Given the lasting consequences of gender disparities for individuals and societies, understanding how early gender differences emerge and to what extent they are driven by broader gender inequalities and norms has become an international priority for research (Ertem et al., 2018; Hyde & Mertz, 2009; Kennedy et al., 2020) and policy (Darmstadt et al., 2019; Gupta et al., 2019; United Nations International Children's Emergency Fund [UNICEF], 2016).

Despite this, research on gender disparities in early childhood development (ECD) is still scarce in low- and middle-income countries (LMICs). Only a small number of studies have explored this topic using internationally comparable data to study gender

differences in ECD across the globe (Bornstein, Putnick, Bradley, et al., 2016; Lu et al., 2020; McCoy et al., 2016) or in a specific region of LMICs (Weber et al., 2017). While almost 90% of the world's children under 5 years live in LMICs, most research on gender disparities in ECD comes from high-income countries (HICs; Hyde, 2014; Zell et al., 2015). The relative dearth of research in LMICs raises concerns about issues of generalizability (i.e., external validity) of the existing knowledge base and constitutes a major barrier for global policy efforts to promote gender equity across the lifespan, such as the United Nations' Sustainable Development Goal (SDG) 4 that aims to "ensure inclusive and equitable quality education and promote lifelong

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learning opportunities for all” and “eliminate gender disparities in education” by 2030 (United Nations, 2018).

To address these limitations, this study leverages internationally comparable data to examine global patterns of gender similarity and difference in ECD of 226,980 children between 36 and 59 months of age living in 71 LMICs. We first assess the magnitude and direction of gender differences by country in the odds of being developmentally on track according to UNICEF’s Early Childhood Development Index (ECDI) and its four developmental domains: literacy-numeracy, approaches to learning, social-emotional, and physical skills (Loizillon et al., 2017; UNICEF, 2014). Then, we estimate between-country variation in gender differences and examine the extent to which gender inequality at the country level explains existing heterogeneity between countries.

Methods

Data

Data for this study come from two large-scale international surveys: the UNICEF Multiple Indicators Cluster Survey (MICS) and the Demographic Health Surveys (DHS), which employed the same questionnaires to assess ECD in multiple study waves. The MICS and DHS surveys provide nationally representative and internationally comparable data about children and their primary caregivers in LMICs. Both surveys use a multistage sample design. In the first stage, primary sampling units (PSUs) are defined, typically census enumeration areas, and selected using probability proportionate to size. In the second stage, households in the selected PSUs are listed, and then selected using random systematic sampling to form the survey clusters (Khan & Hancioglu, 2019). Countries collect these data in face-to-face interviews with respondents answering a standard set of questionnaires that are customized to meet language needs and, in some cases, adding or deleting items specific to individual country needs.

For this study, a total of 71 country surveys implemented between 2010 and 2019 (waves 4, 5, and 6 in MICS) with the most up-to-date available information on early childhood measures and covariates were included in the analyses. The analytic sample for this study comprised 226,980 children aged 36 to 59 months (49% female), which is the period of development for which the ECDI was designed. No participants were excluded, instead data were imputed for participants with missing values in any of the variables of interest. Table 1 provides descriptive statistics for the sample showing the geographic coverage of the sample across world regions. All data used are in the public domain and fully de-identified, therefore this study was given an exempt Institutional Review Board (IRB) review.

Measures

Early Childhood Development. The ECDI is the first population-based measure of ECD that was made available at an internationally comparative level (Loizillon et al., 2017). The ECDI is a caregiver-reported assessment comprised of 10 yes/no questions used to assess children’s abilities to execute developmentally appropriate tasks. ECDI items were developed by child development experts through multistage, multi-country pilot

tests and were validated alongside other tools such as the Early Development Instrument (Janus & Offord, 2007), and the Strengths and Difficulties Questionnaire (Goodman, 2001). Four underlying domains of development that comprised the ECDI were identified through confirmatory factor analyses: literacy-numeracy, approaches to learning, social-emotional, and physical skills (UNICEF, 2014). Each domain was comprised of two or three items and, following UNICEF MICS guidelines, children were considered developmentally “on track” if they exhibited at least 50% of the relevant skills in that domain (Loizillon et al., 2017). For instance, the learning approaches domain included an item on whether the child could follow a simple direction and one item on whether the child could do a task independently; if the child could do at least one of these activities (50%), she was considered developmentally on track. In turn, children who were on track in at least three of the four domains were also coded as developmentally on track according to the overall ECDI. Supplementary Table S1 shows the ECDI items in each developmental domain. For this study, we focused on both the overall ECDI and the four developmental domains.

Country-Level Characteristics. Information from country-level gender equality was obtained from different sources. The Gender Development Index (GDI) and the Gender Inequality Index (GII) were obtained from the United Nations Development Program’s (n.d.) Human Development Data. The GDI female to male ratio measures the gender inequalities in the three dimensions of the Human Development Index (HDI; i.e., a long and healthy life, being knowledgeable, and having a decent standard of living). Values closer to 1 indicate greater equality between women and men, whereas values closer to 0 indicate greater inequality. GII measures potential loss of human development due to gender inequalities in three dimensions: reproductive health (i.e., maternal mortality and adolescent birth rates), women’s empowerment (i.e., proportion of women with at least secondary education and proportion of parliamentary seats held by women), and women’s labor market participation. The index ranges from 0, denoting absolute gender parity, to 1, where females fare as poorly as possible in all dimensions compared to males. We used the GDI and GII information for the corresponding year of data collection in each country. We used multiple imputation to address the fact that five countries were missing GDI data, using HDI and GDP as country-level predictors in the imputation model. When no GII data were available (12 cases), we replaced it with the average value for the country’s region.

Statistical Analysis

Logistic regression models were estimated, by country, using sample weights and clustered standard errors to compute the ratio of the odds for girls to the odds for boys (odds ratios [ORs] and 95% confidence intervals [CIs]) of being developmentally on track according to the ECDI. The same procedure was used for estimating country-level odds ratios for the four ECDI domains. The outcome variable was the binary indicator for being developmentally on track according to the ECDI. The variable sex, which we used as a proxy of gender, was coded 1 for females and 0 for males; thus, odds ratios greater than one indicated a female advantage and odds ratios less than one indicated a male

Table 1. Descriptive Statistics of the Analytic Sample.

Country (region)	Survey	Year	N	Age in months	Female, %	Urban, %
East Asia and Pacific						
Cambodia	DHS	2014	1,700	47.67	49.8	25.9
Kiribati	MICS 6	2019	846	47.49	49.3	36.5
Laos	MICS 6	2017	4,775	46.96	49.5	26.8
Mongolia	MICS 6	2018	2,577	47.44	47.8	50.5
Thailand	MICS 6	2019	5,787	46.88	49.0	35.3
Timor-Leste	DHS	2016	2,696	47.13	48.3	29.7
Tonga	MICS 6	2019	579	47.72	46.3	23.5
Viet Nam	MICS 4	2010	1,463	46.87	49.8	37.6
Europe and Central Asia						
Belarus	MICS 4	2012	1,412	47.70	50.6	70.6
Bosnia and Herzegovina	MICS 4	2011	1,031	46.90	51.5	34.1
Georgia	MICS 6	2018	1,097	47.18	51.0	47.1
Kazakhstan	MICS 5	2015	2,277	47.61	49.2	54.5
Kosovo	MICS 6	2019	621	47.66	48.1	44.0
Kyrgyzstan	MICS 6	2018	1,446	46.94	50.9	37.6
Montenegro	MICS 6	2019	492	47.53	44.1	63.2
North Macedonia	MICS 6	2018	645	47.30	45.0	52.9
Republic of Moldova	MICS 4	2012	733	47.37	46.7	56.8
Serbia	MICS 6	2019	762	47.05	50.1	60.1
Turkmenistan	MICS 6	2019	1,571	47.57	51.8	45.4
Ukraine	MICS 4	2012	1,929	47.34	49.4	63.1
Latin America and the Caribbean						
Argentina	MICS 4	2011	3,625	47.29	48.5	100.0
Belize	MICS 5	2015	1,116	47.47	48.9	41.7
Costa Rica	MICS 6	2018	1,542	47.48	47.7	57.7
Cuba	MICS 6	2019	2,326	46.92	50.6	69.0
Dominican Republic	MICS 5	2014	8,039	47.89	49.9	64.5
El Salvador	MICS 5	2014	3,049	47.86	49.5	55.7
Guyana	MICS 5	2014	1,242	47.59	48.4	20.9
Haiti	DHS	2016	2,352	47.60	49.6	31.0
Honduras	DHS	2011	3,831	47.28	47.6	33.0
Jamaica	MICS 4	2011	671	47.65	47.4	59.2
Mexico	MICS 5	2015	3,417	47.79	49.0	64.2
Paraguay	MICS 5	2016	1,861	47.78	48.2	54.5
Saint Lucia	MICS 4	2012	122	47.67	50.8	36.1
Suriname	MICS 6	2018	1,781	47.40	48.2	55.1
Middle East and North Africa						
Algeria	MICS 6	2018	6,163	47.32	48.7	62.6
Iraq	MICS 6	2018	7,014	46.85	49.3	62.8
Jordan	DHS	2012	2,744	47.49	49.0	71.0
Lebanon ^a	MICS 4	2011	705	47.42	46.8	63.0
Palestine	MICS 6	2019	2,442	46.97	48.3	75.1
Tunisia	MICS 6	2018	1,504	47.73	49.9	63.0
South Asia						
Bangladesh	MICS 6	2019	9,442	47.26	48.4	18.6
Bhutan	MICS 4	2010	2,422	46.67	49.0	21.9
Maldives	DHS	2016	1,283	47.16	47.9	8.7
Nepal	MICS 6	2019	2,886	47.26	48.3	56.3
Pakistan ^a	MICS 4	2010	4,980	47.39	45.2	21.3
Sub-Saharan Africa						
Benin	MICS	2014	4,880	47.22	50.7	49.9
Burundi	DHS	2016	4,844	47.88	50.1	15.4
Cameroon	MICS 5	2014	2,846	46.63	50.8	45.4
Central African Republic	MICS 6	2018	3,703	46.80	50.4	35.3
Chad	MICS 6	2019	9,698	46.47	49.5	17.7

(continued)

Table 1. (continued)

Country (region)	Survey	Year	N	Age in months	Female, %	Urban, %
Congo	MICS 5	2014	3,675	47.95	48.5	30.1
Côte d'Ivoire	MICS 5	2016	3,730	46.63	48.9	25.9
Congo DR	MICS 6	2017	8,704	46.73	51.5	26.8
Eswatini	MICS 5	2014	1,091	47.31	48.9	14.6
Gambia	MICS 6	2018	4,304	47.25	48.8	36.5
Ghana	MICS 6	2018	3,668	47.25	50.7	40.0
Guinea	MICS 5	2016	3,164	47.27	48.4	29.7
Guinea-Bissau	MICS 6	2018	3,130	47.97	48.2	22.3
Lesotho	MICS 6	2018	1,326	47.73	50.0	24.7
Madagascar	MICS 6	2018	5,142	47.26	49.6	22.7
Malawi	MICS 5	2013	7,839	47.17	50.1	11.0
Mali	MICS 5	2015	6,550	46.21	48.6	21.6
Mauritania	MICS 5	2015	4,446	46.73	50.5	42.0
Nigeria	MICS 5	2016	11,648	47.10	49.3	26.5
Rwanda	DHS	2014	1,334	46.73	48.4	19.9
Sao Tome and Principe	MICS 6	2019	756	47.57	49.2	58.2
Senegal ^a	DHS	2017	4,557	47.36	48.3	34.5
Sierra Leone	MICS 6	2017	4,810	47.13	50.0	29.0
Togo	MICS 6	2017	1,989	47.25	47.1	30.2
Uganda	DHS	2016	5,636	47.39	50.5	17.9
Zimbabwe	MICS 6	2019	2,512	47.63	50.0	29.8

Note. DHS: demographic health survey; MICS: multiple indicators cluster survey; DR: democratic republic. This table includes geographical location (continent), the survey and year of data collection, the number of children per country (N), children's average age in months, and the proportion of children who are female and live in an urban area.

^aSome samples cover only a certain geographical area within the country (the Sindh province in Pakistan and Dakar in Senegal) or a specific population group (i.e., Palestinians in Lebanon) and, thus, cannot be interpreted nationally.

advantage. All models were controlled for by children's age to account for differences that are exclusively due to the developmental stage of children. Next, pooled-odds ratios—across all country-level models—were estimated, with attention to heterogeneity across countries. We estimated a pooled-odds ratio for the overall ECDI, and by world regions and ECDI domains. Heterogeneity was calculated using I^2 , where a value of 0% indicated no observed heterogeneity across countries, between 25% and 50% indicated moderate, and > 50% indicated high heterogeneity (Riley et al., 2011). Finally, random-effects meta-regressions were used to assess whether gender inequality indices (i.e., GDI and the GII), explained heterogeneity in gender differences. Only bivariate meta-regressions were tested to avoid multicollinearity between the two gender indexes ($r = .73$).

Across all countries and participants, less than 7% of data on outcome variables were missing, with only seven countries having their percentage of missing values above 10%. Nonetheless, to reduce the risk of bias due to missing data, all analyses were run using multiple imputation (10 imputed data sets) with chained equations for missing values (White et al., 2011). Statistical analyses were conducted using Stata/MP 17.1.

Results

Logistic Models

In half of the countries ($N = 36$), girls and boys displayed no statistically significant differences in the odds of being developmentally on track according to the ECDI. In the other half of the

countries, where differences were evident, these always showed an advantage for girls compared to boys. No differences favoring boys were found in any of the countries included in the sample. As presented in Figure 1, significant odds ratios, favoring girls, ranged from OR = 1.12, 95% CI = [1.01–1.24], $p = .033$, $d = 0.06$ in Nigeria to 2.51, [1.38–4.59], $p = .003$, $d = 0.50$ in North Macedonia. Regarding the four ECDI domains, in 56% ($N = 40$) of the countries, girls' odds of being developmentally on track according to the social-motional domain were statistically significantly higher than the odds for boys. Again, the smallest difference was observed in Nigeria (1.16, [1.04–1.28], $p = .004$, $d = 0.08$) and the largest in North Macedonia (2.20, [1.19–4.07], $p = .012$, $d = 0.44$). In 25% ($N = 18$) of the countries, the odds of being developmentally on track according to the literacy-numeracy domain was also statistically significantly higher than 1 showing a girls' advantage. The differences ranged from 1.17, [1.05–1.30], $p = .004$, $d = 0.09$ in Bangladesh to 2.51, [1.67–3.76], $p < .001$, $d = 0.51$ in Guinea-Bissau. In the approaches to learning domain, there were country-level differences in 8.5% ($N = 6$) of the countries also showing an advantage for girls. No differences were found in the physical health domain for any country (see Figure 2 for country-level odds ratios; odds ratios and 95% CIs for the four ECDI domains are presented in the Supplementary Materials, Table S2).

Meta-Analyses

For the overall ECDI, the random-effects meta-analysis showed a statistically significant pooled-odds ratio that was greater than

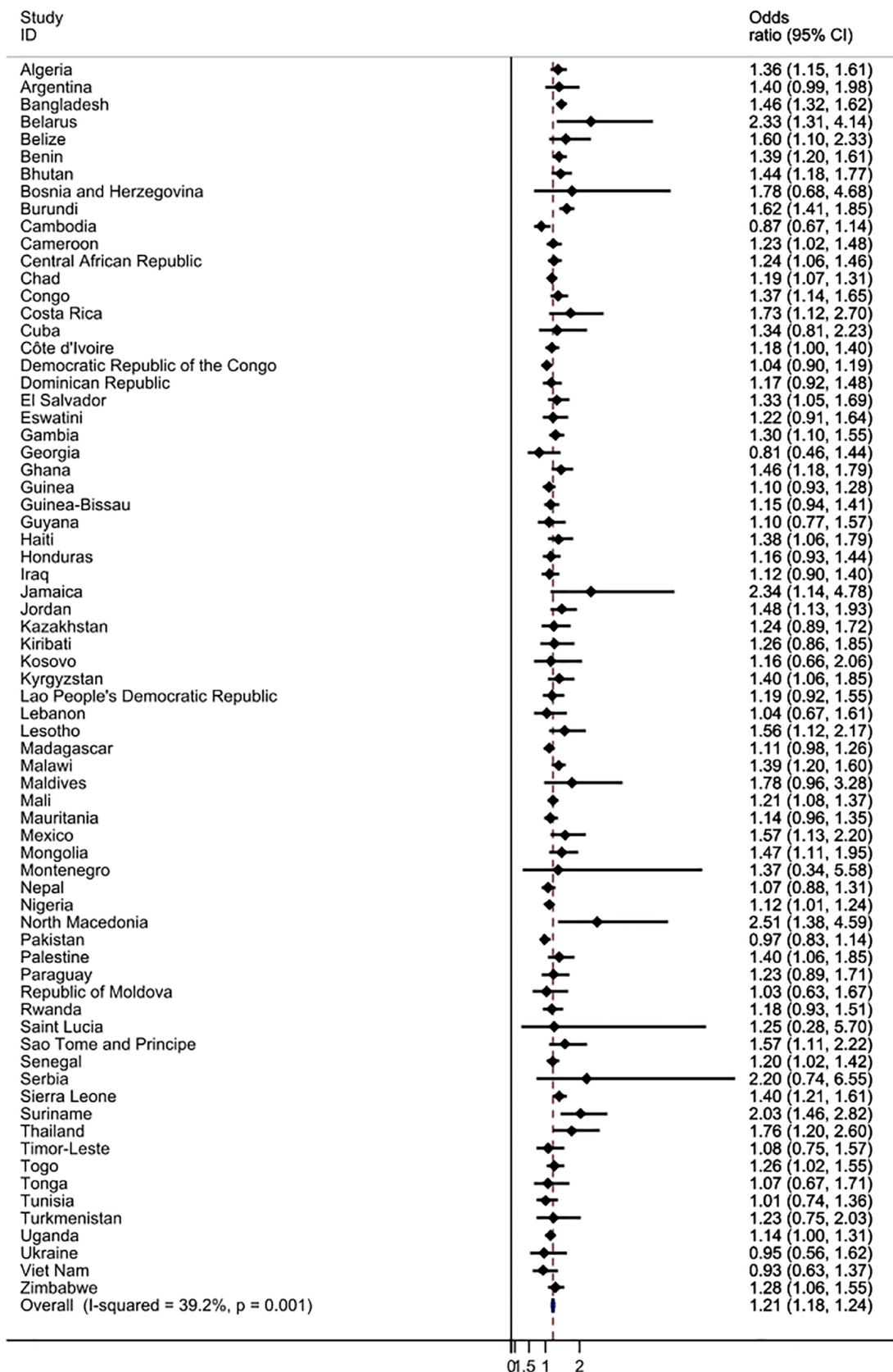


Figure 1. Country Ratios of the Odds for Girls to the Odds for Boys of Being Developmentally on Track According to the Early Childhood Development Index. The analytic sample used for this figure was $N=226,980$ children aged 36 to 59 months (49% female).

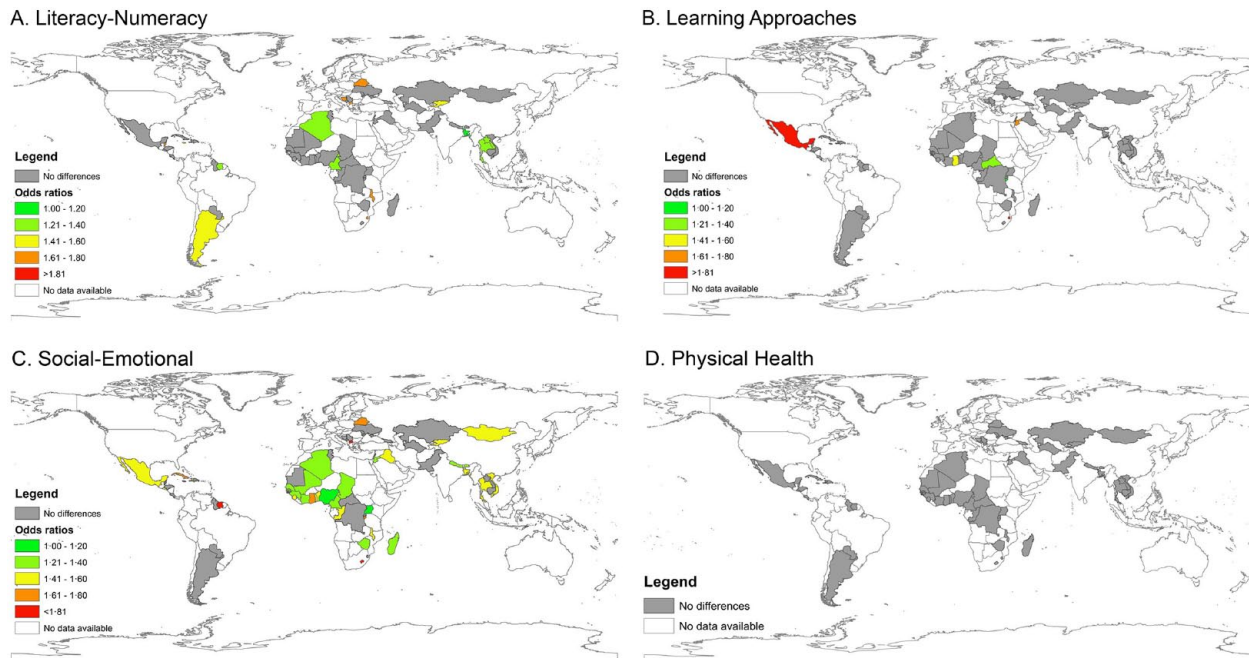


Figure 2. Country Ratios of the Odds for Girls to the Odds for Boys of Being Developmentally on Track According to the Four Early Childhood Development Index Domains. The analytic sample used for this table was $N=226,980$ children aged 36 to 59 months (49% female).

Table 2. Pooled-Odds Ratios of the Odds for Girls to Boys of Being Developmentally on Track According to the ECDI by World Regions and by ECDI Domains.

Region	Odds ratio (95% CI)	p-value	I^2 , %
ECDI pooled-odds ratio (overall ECDI)	1.21 (1.18–1.24)	<.001	39.2
ECDI pooled-odds ratio by world regions			
East Asia Pacific	1.09 (0.96–1.22)	.108	40.6
Europe and Central Asia	1.19 (1.01–1.38)	.483	0.0
Latin America and the Caribbean	1.30 (1.19–1.42)	.509	0.0
Middle-East and North Africa	1.23 (1.11–1.36)	.239	26.1
South Asia	1.22 (1.13–1.31)	<.001	83.2
Sub-Saharan Africa	1.21 (1.17–1.25)	.010	43.6
ECDI pooled-odds ratio by domains			
Literacy-Numeracy	1.13 (1.10–1.17)	.034	24.9
Approaches to Learning	1.05 (1.01–1.08)	.683	0.0
Social-emotional	1.27 (1.24–1.30)	<.001	43.1
Physical	0.98 (0.92–1.03)	.999	11.7

Note. CI: confidence interval; ECDI: Early Childhood Development Index. An odds ratio greater than 1 indicates a girl advantage, an odds ratio of less than 1 would indicate a boy advantage. Estimates were controlled for by children's age to account for differences that are exclusively due to the developmental stage of children. The analytic sample used for this table was $N=226,980$ children aged 36 to 59 months (49% female).

1 indicating a girls' advantage in the odds of being developmentally on track ($OR = 1.21$, 95% $CI = [1.18-1.24]$, $p < .000$; see Table 2). When estimating the pooled-odds ratios by world regions, differences favoring girls were found only in South Asia (1.22, [1.13–1.31], $p < .001$) and Sub-Saharan Africa (1.21, [1.17–1.25], $p = .010$) with substantial heterogeneity identified in the former regions ($I^2 = 83.2\%$) and moderate heterogeneity in the latter ($I^2 = 43.6\%$). No differences were found in the other four regions (see Table 2).

We then looked at the four ECDI domains as subgroup analyses to explore whether some of them would be driving the heterogeneity found in the ECDI. Only the social-emotional and the literacy-numeracy pooled-odds ratios were statistically significant favoring girls in both cases ($OR = 1.27$, 95% $CI = [1.24-1.30]$, $p < .000$ and 1.13, [1.10–1.17], $p < .034$, respectively). Small to moderate heterogeneity was identified in these two domains ($I^2 = 24.9\%$ in literacy-numeracy and $I^2 = 43.1\%$ in social-emotional), which might be an indicator of the heterogeneity found in the overall ECDI.

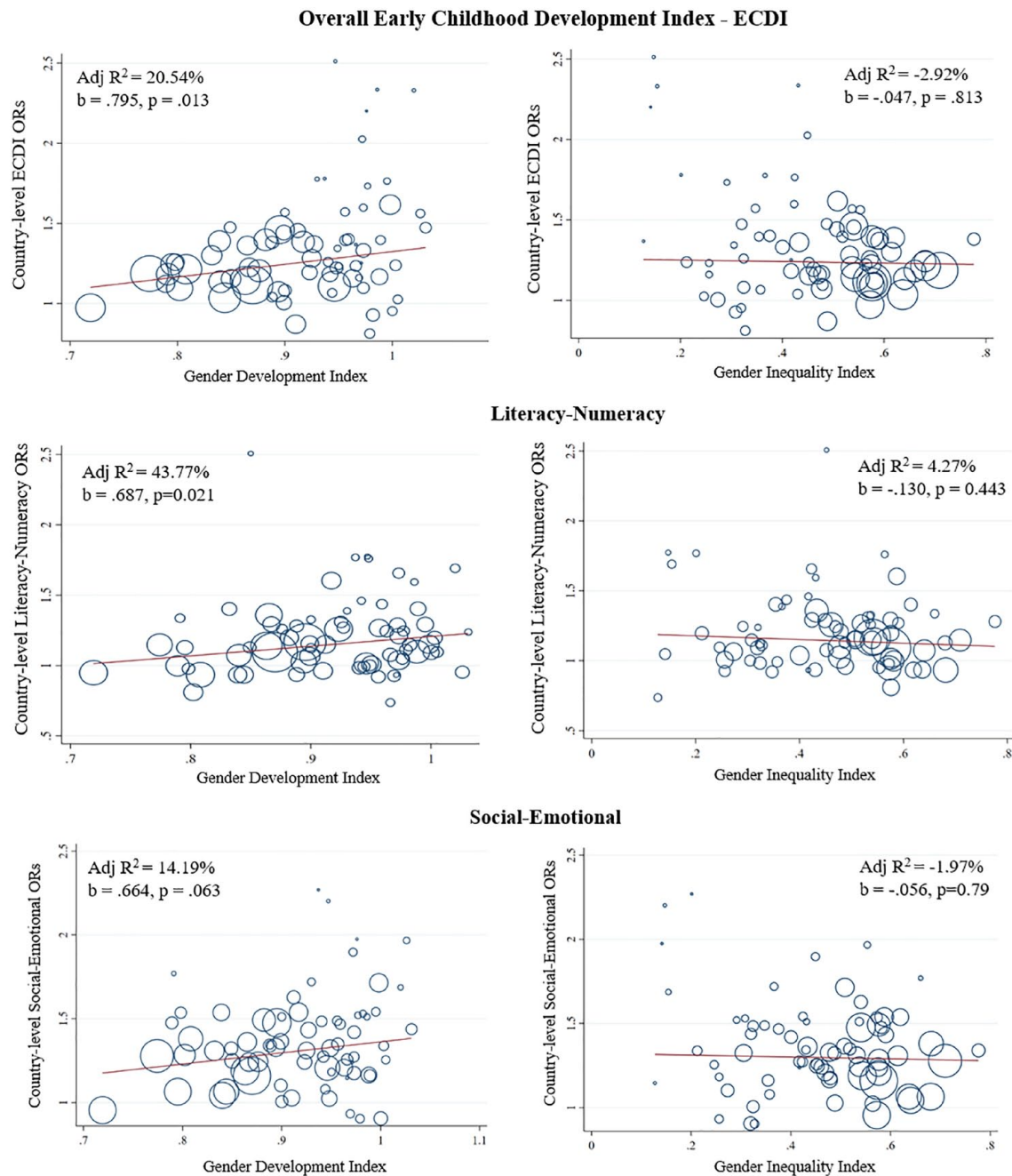


Figure 3. Meta-Regression Models Between Country-Level Odds Ratios and Country-Level Gender Developmental Index and Gender Inequality Index. Given that the adjusted R -squared approximates the percentage of variance explained, if the R -squared is close to zero the adjusted R -squared can be slightly negative. Please interpret the negative values for the adjusted R -squared as an approximation of zero. The analytic sample used for this table was $N = 226,980$. ECDI: Early Childhood Development Index; ORs: odds ratios.

Meta-Regressions

Meta-regressions were estimated only for the domains in which the I^2 was above 40%, which suggested the existence of considerable heterogeneity between countries. Meta-regressions between country-level odds ratios and the GDI explained a proportion of the between-country heterogeneity found in the four remaining

subgroups that were analyzed. In particular, the GDI explained an important amount of the variation between countries in the overall ECDI (20.54%). Regarding the ECDI domains, the GDI explained 43.77% of the heterogeneity between countries in the literacy-numeracy domain and 14.19% of the heterogeneity in the social-emotional domain (see Figure 3). The GII did not explain the heterogeneity found in any of the subgroups analyzed.

Discussion

In half of the countries ($N=36$), the odds of being developmentally on track according to the ECDI were the same for girls and boys. In the other half, however, the ratio of the odds for being developmentally on track was higher than one indicating differences that favored girls. Moreover, it is worth highlighting that even when statistically significant, effect sizes were small to moderate ranging from 0.06 to 0.50. These results are consistent with recent research in LMICs that have also used the ECDI to study ECD. In a cross-sectional study using 60 MICS conducted between 2010 and 2018, researchers found gender statistically significant differences in the overall ECDI in 29 countries (Lu et al., 2020). These differences were, on average, small and favored girls whenever present. This was also the case in another study that used the ECDI to estimate the prevalence of low development in LMICs. When disaggregating the results by gender, the study found that, on average, it was more common for boys than for girls to display low development (McCoy et al., 2016). Nonetheless, recognizing the multidimensionality of child development, this study also tested whether the observed differences in the overall ECDI could be driven by differences in specific domains.

Most differences between the odds for girls and boys were evident in the social-emotional domain ($N=40$), followed by a smaller number of differences in the literacy-numeracy domain ($N=18$). Previous research in HICs had already documented a female advantage in social and behavioral development as early as toddlerhood (Hill et al., 2006). Moreover, recent research focusing on specific regions of the world (Weber et al., 2017) or using fewer LMICs (McCoy et al., 2016) also found female advantages particularly in early language and social-emotional development.

Despite the existing evidence, it remains a challenge to disentangle nature from nurture. Although our results indicate the early emergence of gender differences in social-emotional development, they could also be indicative of caregiver's gender-differentiated expectations of girls' and boys' behavior when reporting their children's skills. Indeed, there is research that suggests girls might develop faster than boys in terms of social and behavior skills (Ertem et al., 2018), whereas other evidence points out that girls might be socialized toward internalizing rather than externalizing behaviors (Keenan & Shaw, 1997). At the same time, recent studies have shown, parental expectations about concordance of children's behavior with traditional gender roles are often potential drivers of gender disparities in social-emotional or behavioral development (Leaper & Brown, 2018; Mesman & Groeneveld, 2018). There is evidence showing that girls and boys living in LMICs receive differential treatment both in terms of parental stimulation and setting limits on children's behaviors, which could ultimately trigger differences in social-emotional skills and behaviors between girls and boys (Bornstein, Putnick, Deater-Deckard, et al., 2016). Indeed, caregiver's differentiated treatment and expectations for girls and boys could potentially influence how parents report and possibly penalize or reward certain behaviors differently for girls and boys, ultimately explaining the observed differences (Bornstein & Putnick, 2016; Endendijk et al., 2017).

Regarding variation of gender differences across countries, moderate variation between countries was observed in the overall ECDI, in particular, driven by the moderate heterogeneity found in the social-emotional domain. In both cases, a proportion of the heterogeneity between countries was explained by

countries' GDI. These results indicate that countries' gender differences, or lack thereof, in ECD are partly explained by countries' levels of gender equality. Our findings indicate that countries with highest gender development displayed also greater advantages for girls. These results are also consistent with previous studies in HICs examining the links between gender equality and students' academic test scores. Recent research found higher country-level gender equality to be associated with a reduction in the gender gap in mathematics scores but an increase in the reading gap in favor of girls (Guiso et al., 2008). Similar results were also found in recent research with countries in the east Asia-Pacific region, where larger differences in ECD favoring girls were associated with higher national gender equality indexes (Weber et al., 2017).

Collectively, these findings expand research that examines gender similarities and differences in ECD by providing evidence from and international comparisons across 71 LMICs. Specifically, the prior work is extended in three ways. First, this study examines a large sample of countries than much of the past work in this area (Lu et al., 2020; McCoy et al., 2016; Weber et al., 2017), and employs the most recent available data. Second, this study complements the previous knowledge base by including a gender analysis of the four ECDI domains, recognizing the multidimensionality of development. Third, this study is the first, to our knowledge, to study the association between gender differences in ECD and country-level gender inequality indices using comparable data across countries in six regions of the world.

A few limitations to our study deserve attention, in part because they offer important topics for future research. Although the ECDI was designed as a brief tool to track and compare basic skills in early childhood across geographically, socioeconomically, and culturally diverse contexts, there has been ongoing debate on its validity and precision to measure specific developmental subdomains (Cappa et al., 2021; Munoz-Chereau et al., 2021; Richter et al., 2019). Moreover, the ECDI has been criticized because its target age range (i.e., 36–59 months) and four domains were not clearly aligned with the SDG indicators (UNICEF, n.d.). To address these limitations, in 2020, UNICEF released the ECDI 2030, which expanded the age range from 24 to 59 months of age and included 20 items to track three developmental domains (i.e., health, learning, and psychosocial well-being) and 12 sub-domains (e.g., expressive language, gross motor, social skills; UNICEF, 2023). However, to date there are very few MICS with the ECDI 2030, making it impossible for now to conduct a global study.

Another limitation with the ECDI is that it is caregiver-reported. Thus, reported developmental differences may contain bias due to caregiver expectations and beliefs (e.g., gendered cultural norms and expectations). Over the last years, different groups of researchers have developed measurement tools such as the International Development and Early Learning Assessment (IDELA; Pisani et al., 2018), the ECD Assessment Scale-Direct Assessment (ECDAS-DA; Rao et al., 2022; Richards et al., 2023), and the Global Scale for Early Development (GSED; Richter et al., 2019) aiming for internationally comparable measures that directly assess development and are able to capture cultural specificity and the multidimensionality of ECD more rigorously. These data will provide important context for our results. Yet, this is not to understate the value of the caregiver reported data; understanding caregivers' perceptions of their children is of value in its own right, even when those perceptions differ from outside observations (Smetana, 2017). Another limitation is that the report of

national data does not allow the identification of gender inequalities at the subnational level, or an examination of intersecting inequalities related to race, ethnicity, migrant status, or disability, among other factors. Finally, it is also important to acknowledge that for the purpose of this study we refer to gender as a binary variable (i.e., girls and boys). However, we acknowledge that gender is a socially and culturally defined construct shaped by social norms and beliefs, hence, future research might study gender as a continuum of individuals' identities and experiences (Hyde et al., 2019).

Considering the importance of experiences in early childhood, as well as the existing disparities in access to learning opportunities between girls and boys, in 2016, the United Nations included in the SDGs target 4.2, which emphasizes on the importance of "ensuring that all girls and boys have access to quality ECD, care and pre-primary education so that they are ready for primary education" (UNICEF, 2016; United Nations, 2019). Yet, despite the increasing policy and advocacy efforts aimed at global improvements in gender equity since early childhood, achieving the SDGs will not be possible without an increased understanding of the current state of early gender disparities in ECD across the world.

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Author Contributions

C.R.-G. conceptualized and designed the study, carried out the initial data curation and data analyses, drafted the initial article, and critically reviewed and revised the article. A.K.Y. critically reviewed and revised the article for important intellectual content. E.D. designed the study, supervised the data curation and analyses, drafted the initial article, and critically reviewed and revised the article. All authors approved the final article as submitted and agree to be accountable for all aspects of the work.

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Data Availability Statement

All data for this study were taken from publicly available data sources. In particular, all MICS files are available from UNICEF's online database at <http://mics.unicef.org/surveys>. All DHS files are available from the DHS online database at <http://www.dhsprogram.com/Data/>. Data for the Gender Inequality Index (GII) is available from the UN Development Program's Human development reports at <http://hdr.undp.org/en/content/gender-inequality-index-gii>.

Supplemental Material

Supplemental material for this article is available online.

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